

SNF-Bench: Separating Static Drift from Natural Flow in Long-Horizon Fixed-Camera Video Generation

Supplementary Material

This supplement records the material supporting the main paper without being required to follow its argument: the region-partition procedure and its sensitivity study, complete audit tables for both tracks at every horizon, measurement coverage, protocols for the two complementary perceptual checks, and additional qualitative material. The text-conditioned and image-conditioned tracks are reported here at equal depth. Section numbering is independent of the main paper.

1. Formal Factor Definitions

The definitions below are those referenced from the main paper’s protocol section. Notation follows it: I_t is frame t of a T -frame sequence, Ω_{static} and Ω_{flow} are the region masks, $\mathbf{u}_t$ is the optical flow from I_t to I_{t+1} , and $\mathcal{E}$ and $\mathcal{L}$ are the early and late windows.

Feature-aligned Background Drift. For matched ORB pairs $\mathcal{M}_t$ between the first frame and frame t ,

$$D_{\text{feat}}(t) = \text{median}_{(p,q) \in \mathcal{M}_t} \|p - q\|_2, \\ \text{fBD} = \frac{100}{d} \cdot \frac{1}{|\mathcal{L}|} \sum_{t \in \mathcal{L}} D_{\text{feat}}(t) \quad (\downarrow), \quad (1)$$

with d the frame diagonal, so fBD is a percentage of image size.

Normalised Background Flow.

$$\text{NBF} = \frac{10^3}{W \Delta t} \cdot \frac{1}{T-1} \sum_{t=1}^{T-1} \frac{1}{|\Omega_{\text{static}}|} \sum_{\mathbf{x} \in \Omega_{\text{static}}} \|\mathbf{u}_t(\mathbf{x})\|_2 \quad (\downarrow), \quad (2)$$

with W the frame width and Δt the interval of the sampled pairs actually used for flow estimation, so the quantity is independent of both the native frame rate and the sampling policy.

Global-motion compensation. With $T_t(\mathbf{x}) = s_t R_t \mathbf{x} + \mathbf{b}_t$ the robust similarity transform fitted to static-region correspondences and $\mathbf{d}_t(\mathbf{x}) = T_t(\mathbf{x}) - \mathbf{x}$ the displacement field it induces,

$$\tilde{\mathbf{u}}_t(\mathbf{x}) = \mathbf{u}_t(\mathbf{x}) - \mathbf{d}_t(\mathbf{x}). \quad (3)$$

Motion-Compensated Foreground Flow and Flow Persistence.

$$\text{MCFF}(\mathcal{W}) = \frac{1}{|\mathcal{W}|} \sum_{t \in \mathcal{W}} \frac{1}{|\Omega_{\text{flow}}|} \sum_{\mathbf{x} \in \Omega_{\text{flow}}} \|\tilde{\mathbf{u}}_t(\mathbf{x})\|_2 \quad (\uparrow), \quad (4)$$

$$\text{FP} = \min \left(\frac{\text{MCFF}(\mathcal{L})}{\text{MCFF}(\mathcal{E}) + \epsilon}, c \right) \quad (\uparrow), \quad (5)$$

clipped at a fixed c so that a vanishing early window cannot produce an unbounded ratio.

Drift leakage and attenuation. With F_{static} the mean flow magnitude over Ω_{static} and $F_{\text{raw}}, F_{\text{comp}}$ the mean raw and compensated magnitudes over Ω_{flow} , all in the late window,

$$\text{DLR} = \frac{F_{\text{static}}}{F_{\text{raw}} + \epsilon} \quad (\downarrow), \quad \text{DAR} = 1 - \frac{F_{\text{comp}}}{F_{\text{raw}} + \epsilon} \quad (\downarrow). \quad (6)$$

DLR is not a fraction and is deliberately unbounded: $\text{DLR} > 1$ indicates static support moving more than the region intended to move. DAR is stored signed and reported clipped to $[0, 1]$.

2. Region Partition

Derivation. For every generated sequence we average optical-flow magnitude [7] over the leading 12% of sampled frame pairs and threshold it by Otsu’s method [4] into a dynamic region and its complement. The static region is then reduced by a fixed morphological erosion and a 4% frame border discarded; together these form the ignored transition band, so boundary pixels—where the partition is least certain and flow least reliable—contribute to neither region. The same procedure applies to both tracks.

Why the estimate is restricted to an early window. Deriving a partition from a sequence’s entire motion field would be circular, since accumulated background drift would itself be read as evidence that the affected region is dynamic. Restricting the estimate to the early window breaks that circularity in the direction that matters: at the 60s horizon the partition is fixed from the first 7.1s, before long-horizon drift has accumulated, and the metrics are evaluated over the remainder.

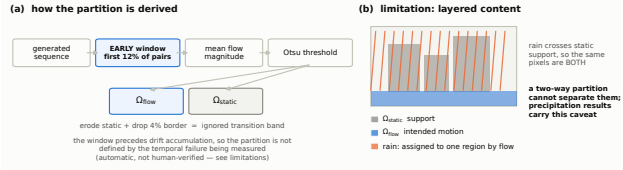


Figure 1. Region-partition protocol. (a) The partition is derived automatically from early-window flow, with an ignored transition band at the boundary. (b) A two-way partition cannot represent layered content such as rain crossing static support, which is the scope limit stated in the main paper.

Limitations. The partition is obtained automatically rather than by annotation, so no inter-annotator agreement is reported, and for the image-conditioned track it is derived per generated sequence rather than once from the shared source image. The early window is a weaker guarantee than a first-frame annotation: a system that drifts within the first 12% of its rollout can influence its own partition. We therefore treat partitions as measured quantities and report the consequence directly—every measurement is recomputed under systematic erosion and dilation of the boundary, with the response in Table 1.

3. Validation Protocol

Geometric severity is specified in a common physical unit—the mean pixel displacement a corruption induces over the frame by the end of the rollout—and each family’s parameter is derived from it using the frame geometry: a translation of d pixels, a rotation of $d/\bar{r}$ radians, or a zoom of $1 + d/\bar{r}$, where $\bar{r}$ is the mean distance from the frame centre. This is what makes the geometric families comparable. A parameter in degrees is not commensurate with one in pixels: at the evaluation resolution one degree of rotation induces roughly 2.7 px of mean displacement, so families compared by their native parameters differ in injected severity by more than an order of magnitude, and a factor can appear non-monotone simply because the corruption was too weak to clear the clip’s own baseline drift.

4. Generation Configuration

Every audited system was run in its released configuration. What the run manifest records is reported below; what it does not record is stated rather than reconstructed. Per-method sampler settings—denoising steps, chunk length, guidance scale, cache policy—were not captured at generation time, and we do not infer them after the fact. This is the weakest point in the reproducibility of the audit, and it constrains what the

Factor	target	ρ	ρ_{rot}	off-target	admitted
fBD	drift	+1.00	+0.70	14%	✓
NBF	drift	+1.00	+0.60	6%	✓
MCFF-L	freeze	+1.00	n/a	14%	✓
FP	freeze	+0.80	n/a	21%	✓
DLR	drift	+1.00	-0.80	5%	✓
DAR	drift	+0.90	+1.00	18%	✓
VBench DD	drift	+0.70	+0.70	2%	✓

Table 1. Mechanistic validation and factor admission. Each factor is scored on the corruption its definition says it should detect: accumulating global drift for the static-fidelity and drift diagnostics, progressive freezing for the persistence factors. ρ is Spearman correlation against injected severity, signed so that a correct response is positive; ρ_{rot} repeats it for rotational drift, the harder case. off-target is the largest relative excursion under corruptions the factor should ignore—photometric drift and perturbation of the region partition—so a small value indicates selectivity. Admission requires $\rho \geq 0.7$ and off-target response below 25%. Twelve reference clips per severity level.

results support: they compare the systems as released and run here, not the space of settings each admits.

5. Measurement Coverage

Coverage is counted from valid per-video measurements rather than from the presence of a results file, so a run that terminated early is reported as partial. Separately, fBD is feature-based [6] and declines to report where too few repeatable keypoints survive; those abstentions are tabulated rather than folded silently into a smaller n .

6. Complete Audit, Both Tracks

The main paper reports the text-conditioned track at 60s, the tier whose sample supports method-level separation. Both tracks appear here at every horizon, with uncertainty from a percentile bootstrap over items [1]. For the image-conditioned track, systems marked matched run under a common long-horizon wrapper rather than their released configuration, so their scores are read as deployment sensitivity rather than as the published method’s performance.

7. Aggregation Sensitivity

The evaluation set is dominated by one flow medium, so the audit must be shown not to be a report on that medium alone. The main paper aggregates by averaging over prompts. The alternative, weighting each scene category equally, is not preferable here: at the principal horizon three of the five populated categories contain two prompts each, so a macro-average would give a two-prompt cell the same influence as a fourteen-

System	setting	resolution	fps	horizons
CausVid	native	832×480	16	5s, 60s, 120s, 240s
Self-Forcing	native	832×480	16	5s, 60s, 120s, 240s
Infinite-Forcing	native	832×480	16	5s, 60s, 120s, 240s
Rolling-Forcing	native	832×480	16	5s, 60s, 120s, 240s
Reward-Forcing	native	832×480	16	5s, 60s, 120s, 240s
LongLive	native	832×480	16	5s, 60s, 120s, 240s
Causal-Forcing	native	832×480	16	5s, 60s, 120s, 240s

Table 2. Recorded generation configuration. Every audited system was run in its released configuration and produced output at the same resolution and frame rate over the same horizons, so no comparison in this paper is confounded by output format. Per-method sampler settings—denoising steps, chunk length, guidance scale, cache policy—were not captured in the run manifest and are therefore not listed here; this is a reproducibility gap we state rather than fill by inference.

track	horizon	measured	fBD abstained	rate
I2V	120s	400	0	0.0%
I2V	240s	65	0	0.0%
I2V	5s	180	0	0.0%
I2V	60s	630	7	1.1%
T2V	120s	42	0	0.0%
T2V	240s	28	0	0.0%
T2V	5s	48	0	0.0%
T2V	60s	185	0	0.0%

Table 3. Measurement coverage and fBD abstention. fBD is feature-based and declines to report when too few repeatable keypoints survive in the static region. Every abstention in this benchmark occurs on a single desert dust-storm scene, whose flat, dust-obscured ground offers no stable structure to track, and it occurs there for every system alike—so it reflects the scene, not a method. The remaining factors are reported for those clips as usual; only fBD is withheld.

prompt one and inflate variance in exchange for nominal balance. We therefore report both and state what changes.

The best- and worst-ranked systems are the same under either aggregation, so the contrast the paper draws does not depend on the choice. Intermediate ranks do move, which is why no ordinal claim is made about the middle of the field.

8. Whole-Frame and SNF-Bench Rank Correlation

9. Deployment Sensitivity

10. Signed Drift Attenuation

Drift attenuation is stored signed and reported clipped to $[0, 1]$. Negative values are informative rather than anomalous: where local flow opposes the estimated global field, compensation increases the measured mag-

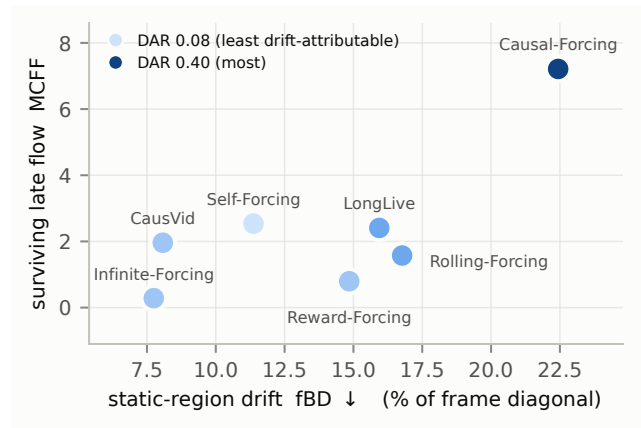


Figure 2. Operating regimes under native deployment. Each system in the static-drift-surviving-flow plane. Colour encodes the relative attenuation of measured dynamic-region flow after global-motion compensation, not a share of it attributable to drift: the quantity is signed, and values are shown clipped to $[0, 1]$. The view exposes trade-offs; it is not a leaderboard.

nitude. We report their incidence rather than suppressing it.

11. Evaluation-Set Composition

12. Complementary Perceptual Checks

Neither check is aggregated into a factor and neither is a preference study. Both answer one question: does an independent judge, with no access to our factors, order the same clip pairs the same way?

Pair selection. Pairs are chosen by rule rather than by eye. For each axis, candidate pairs of systems on a shared prompt are binned by the relative gap in the relevant factor and sampled across three bands (0.05–0.20, 0.20–0.50, 0.50–1.00), so agreement is measured

Method	status	setting	5s	60s	120s	240s
CausVid	public	native	V6 B	V23 T23 B	V6 B	V4 B
Self-Forcing	public	native	V6 B	V23 T23 B	V6 B	V4 B
Infinite-Forcing	public	native	V6 B	V23 T23 B	V6 B	V4 B
Rolling-Forcing	public	native	V6 B	V24 T24 B	V6 B	V4 B
Reward-Forcing	public	native	V12 B	V23 T23 B	V6 B	V4 B
LongLive	public	native	V6 B	V23 T23 B	V6 B	V4 B
Causal-Forcing	public	native	V6 B	V23 T23 B	V6 B	V4 B
Method	status	setting	5s	60s	120s	240s
Causal-Forcing++ (2-step)	public	matched	V10 T10 X B	V30 T30 X B	V20 T20 X B	V5 T5 X B
Causal-Forcing++ (1-step)	public	matched	V10 T10 X B	V30 T29/30 X B	V20 T20 X B	V5 T5 X B
Causal-Forcing (frame-wise)	public	matched	V10 T10 X B	V30 T30 X B	V20 T20 X B	V5 T5 X B
Causal-Forcing++ (2-step, frame-wise)	public	native	V10 T10 X B	V30 T30 X B	V20 T20 X B	V5 T5 X B
Self-Forcing	public	matched	V10 T10 X B	V30 T30 X B	V20 T20 X B	V5 T5 X B
CausVid	public	matched	V10 T10 X B	V30 T29/30 X B	V20 T20 X B	V5 T5 X B
Wan2.1-I2V-14B-480P	public	native	V10 T10 X B	.	.	.
Wan2.2-I2V-A14B	public	native	V10 T10 X B	.	.	.
LTX-Video 13B-0.9.8-distilled	public	native	V10 T10 X B	.	.	.
Causal-Forcing++ (1-step, frame-wise)	public	native	.	V30 T30	V20 T20	.

Table 4. Asset and score coverage for the audited public systems. Tn/N marks entries where a metrics file exists but only n of N videos hold usable measurements.

over the whole operating range rather than only where any metric would look good. Twelve pairs are drawn per axis.

Human study. Raters answer exactly two forced-choice questions—which video’s background moves more, and in which video the moving content slows or stops more—and are never asked which is better, since “better” conflates the two failures the benchmark separates. Trial order and left/right presentation are randomised per rater, choices are resolved back to system identities server-side, and system names are never shown. Ties are permitted: forcing a choice on an indistinguishable pair manufactures noise. Twenty-four trials per rater, about eight minutes, three raters.

Vision–language judge. The same pairs, with the same wording, are put to an open-weight vision–language model [5], following the practice of MLLM-scored generation benchmarks [2, 3] but restricted to the two axis questions rather than an overall score. Each pair is asked twice with the clips swapped; a judge answering by position rather than by content disagrees with itself, so agreement is computed only over self-consistent trials and the self-consistency rate is reported beside it.

Reporting the human study. For the human study we report, per axis, the agreement rate with a binomial interval, the tie rate, and agreement resolved by metric gap. The last matters: a factor that agrees with observers only on large gaps is a weaker result than one that agrees throughout, and pooling hides the difference. The same format is defined for the vision–language judge, but as set out below its trials did not

meet the precondition for reporting an agreement rate at all.

Outcome of the vision–language check: negative. We report this check as a negative result. Before reading any agreement rate we required the judge to pass two controls. Shown one clip against itself, where the only correct answer is that the two are the same, it answered correctly on five of six pairs. Shown a frozen single frame against a real sequence, where the moving sequence must be chosen and the choice must follow the clip when the two are swapped, it succeeded on three of six. On the audited pairs it was self-consistent under swapping on seven of twenty-four, which is what a three-way choice made at random would produce. An agreement rate computed over so few self-consistent trials describes the judge’s presentation bias rather than the videos, so we do not report one, and no number from this check enters any factor, table or figure elsewhere in this work.

We record the outcome because it is informative about the task rather than only about the model. The distinction the benchmark draws—between a background that drifts and intended motion that decays—lies below the discrimination threshold of a current open-weight judge at this scale, even though the same judge reliably detects that a clip is identical to itself. This is consistent with the main finding that whole-frame motion scores do not separate these two failures either, and it is the reason the benchmark rests on mechanistic validation under injected corruptions rather than on a model’s preferences. We report the protocol and its controls in full so the check can be repeated at a larger scale; we do not present it as evidence for the factors.

Method	fBD↓	NBF↓	MCFF-E	MCFF-L↑	FP↑	DLR↓	DAR↓
5s horizon ($n = 6$)							
CausVid	2.08	6.56	1.44	2.91	1.24	0.26	0.07
Self-Forcing	0.68	10.8	7.70	4.35	0.76	0.49	0.03
Infinite-Forcing	0.24	2.32	1.23	0.89	1.08	0.21	0.09
Rolling-Forcing	3.51	12.1	3.10	3.69	1.32	0.48	0.17
Reward-Forcing	1.10	5.21	0.63	0.58	1.37	0.44	0.28
LongLive	1.91	7.31	1.25	2.27	1.16	0.59	0.32
Causal-Forcing	9.55	31.0	5.32	3.01	1.13	0.62	0.37
60s horizon ($n = 23$)							
CausVid	8.08	9.27	2.87	1.96	0.67	0.45	0.14
Self-Forcing	11.4	12.7	10.4	2.54	0.44	0.50	0.14
Infinite-Forcing	7.75	3.65	1.17	0.29	0.63	0.55	0.14
Rolling-Forcing	16.8	11.2	3.18	1.57	0.75	0.54	0.18
Reward-Forcing	14.8	8.51	3.81	0.80	0.49	0.60	0.17
LongLive	15.9	13.9	6.46	2.41	0.75	0.59	0.17
Causal-Forcing	22.4	87.6	9.84	7.21	0.83	0.94	0.43
120s horizon ($n = 6$)							
CausVid	6.15	3.42	0.36	0.52	1.06	0.41	0.13
Self-Forcing	21.2	6.80	4.85	0.82	0.85	0.75	0.15
Infinite-Forcing	16.1	2.60	0.50	0.32	0.64	0.57	0.21
Rolling-Forcing	23.8	10.6	3.33	0.77	0.38	0.82	0.22
Reward-Forcing	20.0	11.6	2.05	0.69	0.39	0.70	0.12
LongLive	22.5	9.66	5.10	1.19	0.81	0.84	0.23
Causal-Forcing	21.8	130	10.5	10.7	0.90	0.81	0.50
240s horizon ($n = 4$)							
CausVid	7.23	13.4	0.42	0.87	0.99	0.85	0.08
Self-Forcing	20.0	17.4	3.09	0.50	0.15	2.26	0.20
Infinite-Forcing	20.0	4.15	0.33	0.23	1.02	1.08	0.15
Rolling-Forcing	23.7	15.4	2.15	0.97	0.46	0.78	0.18
Reward-Forcing	26.4	11.4	1.09	3.55	1.53	0.50	0.07
LongLive	23.1	11.8	1.52	1.32	1.20	1.38	0.31
Causal-Forcing	18.9	102	4.72	6.90	1.21	0.73	0.58

Table 5. Complete T2V audit, all four horizons. The main paper reports the 60s and 240s panels; the 5s tier is an initialisation check and 120s an intermediate diagnostic. Every row is computed under the same frozen specification, so panels are comparable within a horizon; prompt sets differ across horizons, so columns are not comparable between panels. All systems are evaluated in the native setting. n is stated per horizon and is common to every row of that panel.

13. Additional Qualitative Material

References

- [1] Bradley Efron. Bootstrap methods: Another look at the jackknife. *Ann. Statist.*, 7(1):1–26, 1979. 2
- [2] Hui Han, Siyuan Li, Jiaqi Chen, Yiwen Yuan, Yuling Wu, Yufan Deng, Chak Tou Leong, Hanwen Du, Junchen Fu, Youhua Li, Jie Zhang, Chi Zhang, Li-jia Li, and Yongxin Ni. Video-Bench: Human-aligned video generation benchmark. In *CVPR*, pages 18858–18868, 2025. 4
- [3] Xuan He, Dongfu Jiang, Ge Zhang, Max Ku, Achint Soni, Sherman Siu, Haonan Chen, Abhranil Chandra, Ziyang Jiang, Aaran Arulraj, Kai Wang, Quy Duc Do, Yuansheng Ni, Bohan Lyu, Yaswanth Narsupalli,

- Rongqi Fan, Zhiheng Lyu, Bill Yuchen Lin, and Wenhui Chen. VideoScore: Building automatic metrics to simulate fine-grained human feedback for video generation. In *Proc. Conf. Empirical Methods Natural Language Process. (EMNLP)*, pages 2105–2123, 2024. 4
- [4] Nobuyuki Otsu. A threshold selection method from gray-level histograms. *IEEE Trans. Syst. Man Cybern.*, 9(1):62–66, 1979. 1
- [5] Qwen Team. Qwen3-VL technical report. *arXiv preprint arXiv:2511.21631*, 2025. 4
- [6] Ethan Rublee, Vincent Rabaud, Kurt Konolige, and Gary Bradski. ORB: An efficient alternative to SIFT or SURF. In *ICCV*, pages 2564–2571, 2011. 2
- [7] Zachary Teed and Jia Deng. RAFT: Recurrent all-pairs field transforms for optical flow. In *ECCV*, pages 402–419, 2020. 1

Method	setting	fBD↓	NBF↓	MCFF-E	MCFF-L↑	FP↑	DLR↓	DAR↓
5s horizon ($n = 10$)								
Causal-Forcing++ (2-step)	matched	3.69	10.0	–	0.65	0.88	0.67	0.22
Causal-Forcing++ (1-step)	matched	5.28	11.4	–	0.51	0.66	0.66	0.24
Causal-Forcing (frame-wise)	matched	16.1	45.3	–	3.28	0.40	0.68	0.12
Causal-Forcing++ (2-step, frame-wise)	native	1.22	10.6	–	0.83	0.49	0.68	0.11
Self-Forcing	matched	9.66	5.74	–	0.33	0.65	0.74	0.07
CausVid	matched	15.2	10.8	–	0.90	0.95	1.04	0.06
Wan2.1-I2V-14B-480P	native	5.01	21.3	–	4.84	1.03	0.52	0.16
Wan2.2-I2V-A14B	native	6.31	25.3	–	9.08	0.95	0.50	0.26
LTX-Video 13B-0.9.8-distilled	native	6.65	17.8	–	3.68	0.90	0.39	0.18
60s horizon ($n = 30$)								
Causal-Forcing++ (2-step)	matched	9.18	5.97	–	0.67	0.86	0.67	0.15
Causal-Forcing++ (1-step)	matched	5.34	7.05	0.86	0.57	0.77	0.64	0.28
Causal-Forcing (frame-wise)	matched	21.1	36.4	2.80	2.69	0.89	0.80	0.35
Causal-Forcing++ (2-step, frame-wise)	native	9.94	6.91	–	0.81	0.98	0.75	0.11
Self-Forcing	matched	17.3	5.52	–	0.74	0.88	0.67	0.14
CausVid	matched	19.9	3.31	0.58	0.41	0.81	0.57	0.14
Causal-Forcing++ (1-step, frame-wise)	native	10.4	5.25	–	0.52	0.97	0.83	0.14
120s horizon ($n = 20$)								
Causal-Forcing++ (2-step)	matched	9.65	8.81	–	0.93	0.91	0.58	0.14
Causal-Forcing++ (1-step)	matched	7.26	9.39	–	0.83	0.84	0.62	0.16
Causal-Forcing (frame-wise)	matched	19.9	52.7	4.82	5.06	1.08	0.67	0.24
Causal-Forcing++ (2-step, frame-wise)	native	11.4	17.3	–	1.54	0.96	0.75	0.06
Self-Forcing	matched	23.0	10.3	–	0.78	0.44	0.66	0.20
CausVid	matched	19.7	6.04	1.15	0.83	0.67	0.48	0.14
Causal-Forcing++ (1-step, frame-wise)	native	13.6	7.39	–	0.43	0.68	0.77	0.20
240s horizon ($n = 5$)								
Causal-Forcing++ (2-step)	matched	10.4	4.83	–	0.63	1.03	0.55	0.09
Causal-Forcing++ (1-step)	matched	6.42	6.66	0.56	0.51	0.83	0.67	0.30
Causal-Forcing (frame-wise)	matched	19.8	31.5	–	2.86	0.76	0.69	0.33
Causal-Forcing++ (2-step, frame-wise)	native	6.32	3.95	–	0.69	0.66	0.83	0.01
Self-Forcing	matched	22.3	5.74	–	1.66	0.79	0.71	0.05
CausVid	matched	18.4	2.08	–	0.83	0.83	0.44	0.07

Table 6. Complete I2V audit, all four horizons. The main paper reports the 60s and 240s panels; the 5s tier is an initialisation check and 120s an intermediate diagnostic. Every row is computed under the same frozen specification, so panels are comparable within a horizon; prompt sets differ across horizons, so columns are not comparable between panels. n is stated per horizon and is common to every row of that panel.

System	prompt mean	rank	macro mean	rank
NBF				
Infinite-Forcing	3.65	1	4.12	1
Reward-Forcing	8.51	2	10.11	4 [†]
CausVid	9.27	3	9.93	3
Rolling-Forcing	11.23	4	11.32	5 [†]
Self-Forcing	12.72	5	8.98	2 [†]
LongLive	13.86	6	14.69	6
Causal-Forcing	87.62	7	61.70	7
fBD				
Infinite-Forcing	7.75	1	6.10	1
CausVid	8.08	2	9.51	3 [†]
Self-Forcing	11.37	3	9.27	2 [†]
Reward-Forcing	14.85	4	15.61	5 [†]
LongLive	15.94	5	14.93	4 [†]
Rolling-Forcing	16.77	6	18.85	6
Causal-Forcing	22.44	7	23.49	7

Table 7. Sensitivity of the audit to category aggregation. Prompt means, as reported in the main paper, against category macro-averages, for the two static-fidelity factors at the principal horizon. [†] marks a system whose rank changes. The best- and worst-ranked systems are identical under both, so the paper’s headline contrast does not depend on the choice; intermediate positions do change, which is why no ordinal claim is made about the middle of the field.

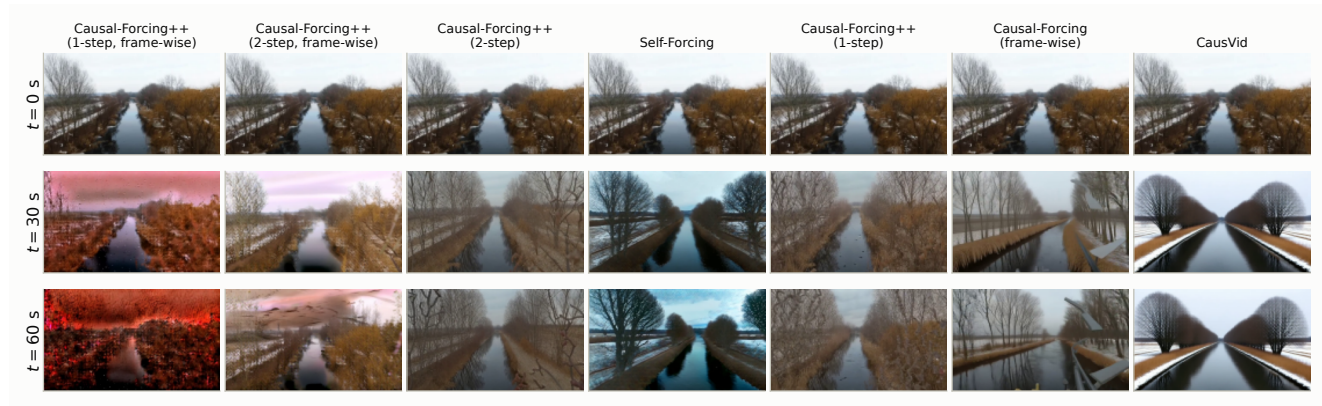


Figure 3. Image-conditioned systems on one shared source image. Every system begins from the identical frame, so divergence over the rollout is attributable to the system rather than to a differently imagined scene—which is not true of the text-conditioned track, where each system authors its own layout. Columns are ordered by increasing static-region drift.

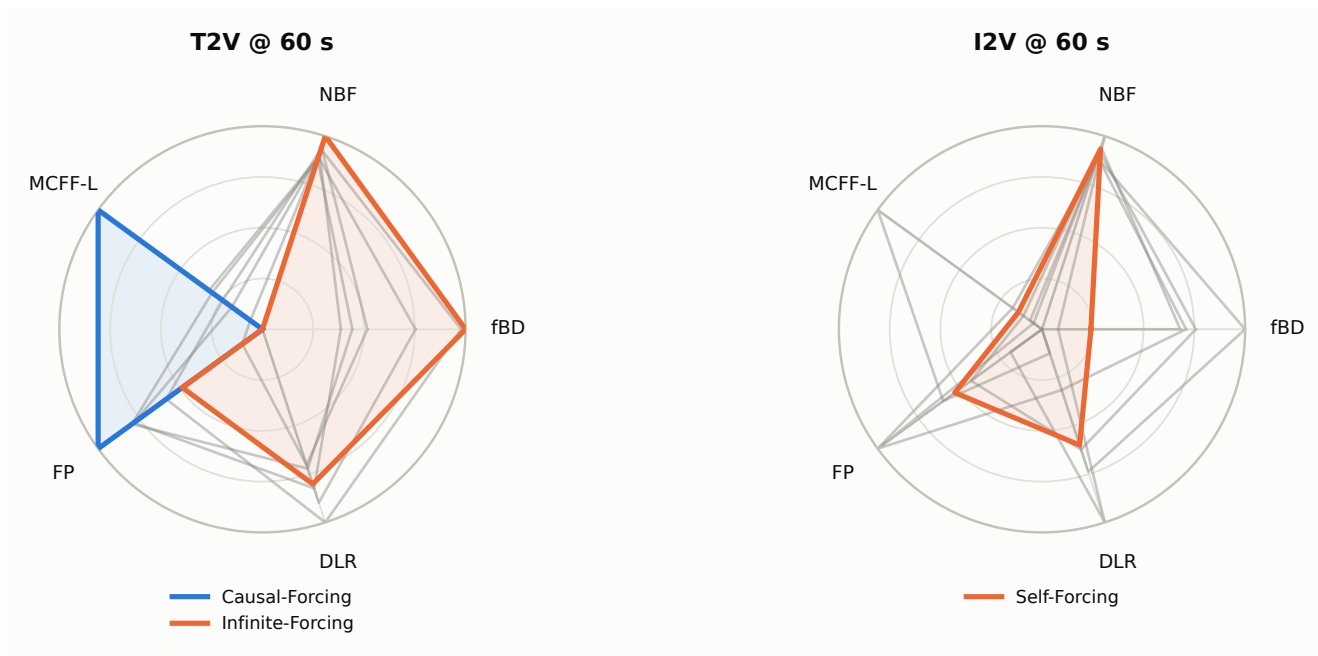


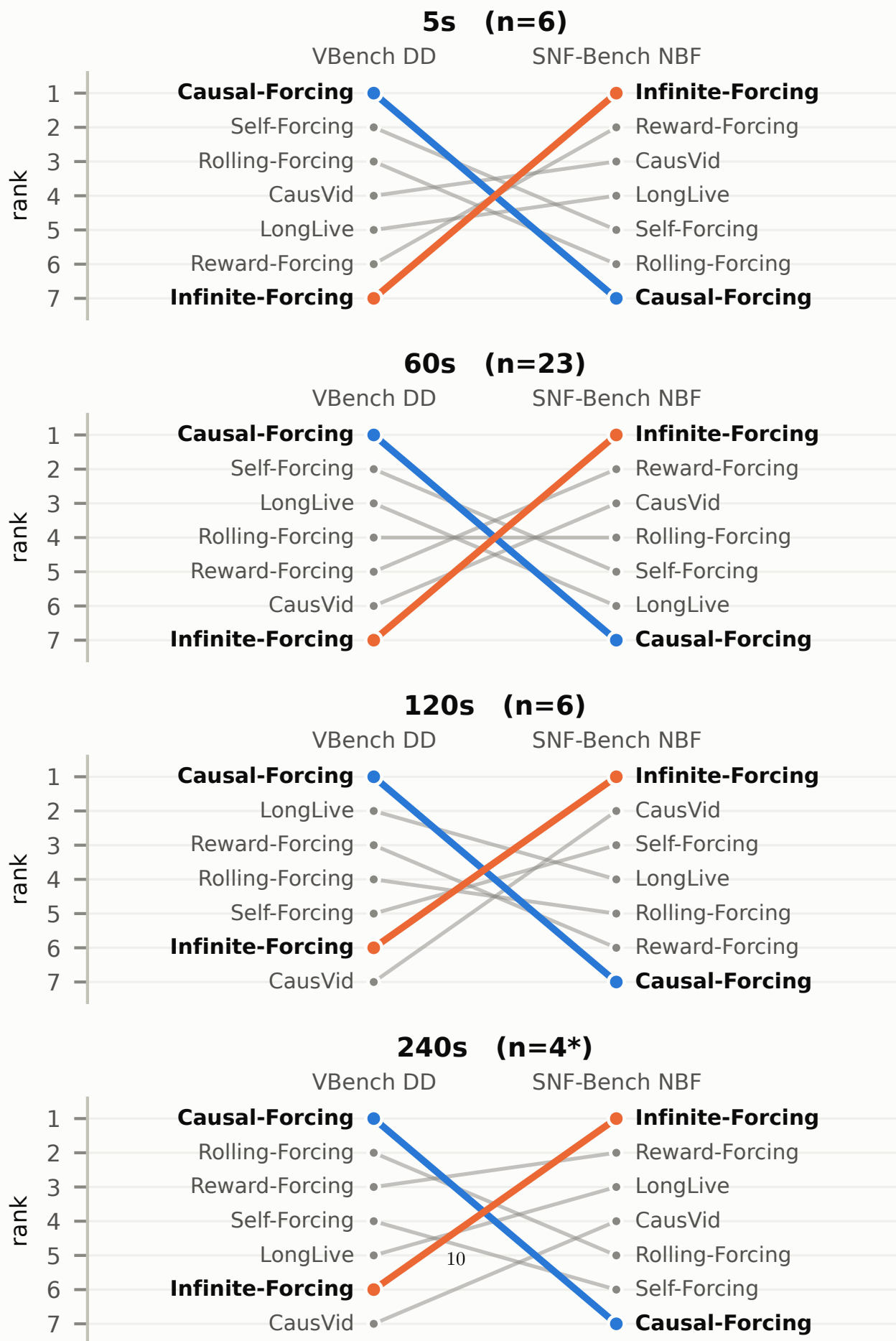
Figure 4. Factor profile per track. Each factor is rescaled across the audited systems and oriented so that outward is always better; plotting raw values would place “most drift” and “most surviving motion” both outward and the shape would carry no meaning. No system encloses the others on either track, which is the conclusion the operating-regime view reaches, shown as a profile.



Figure 5. Stated scope boundaries. (a) A two-way partition cannot separate precipitation from the support it crosses. (b) A system can attain near-best static fidelity precisely because its intended motion has also stopped. (c) High leakage with low attenuation identifies support that moves without translating, which a rigid global model does not explain.



Figure 6. Every audited text-conditioned system on one prompt. Columns are ordered by increasing static-region drift; the two highlighted systems are the pair a whole-frame motion score cannot separate.



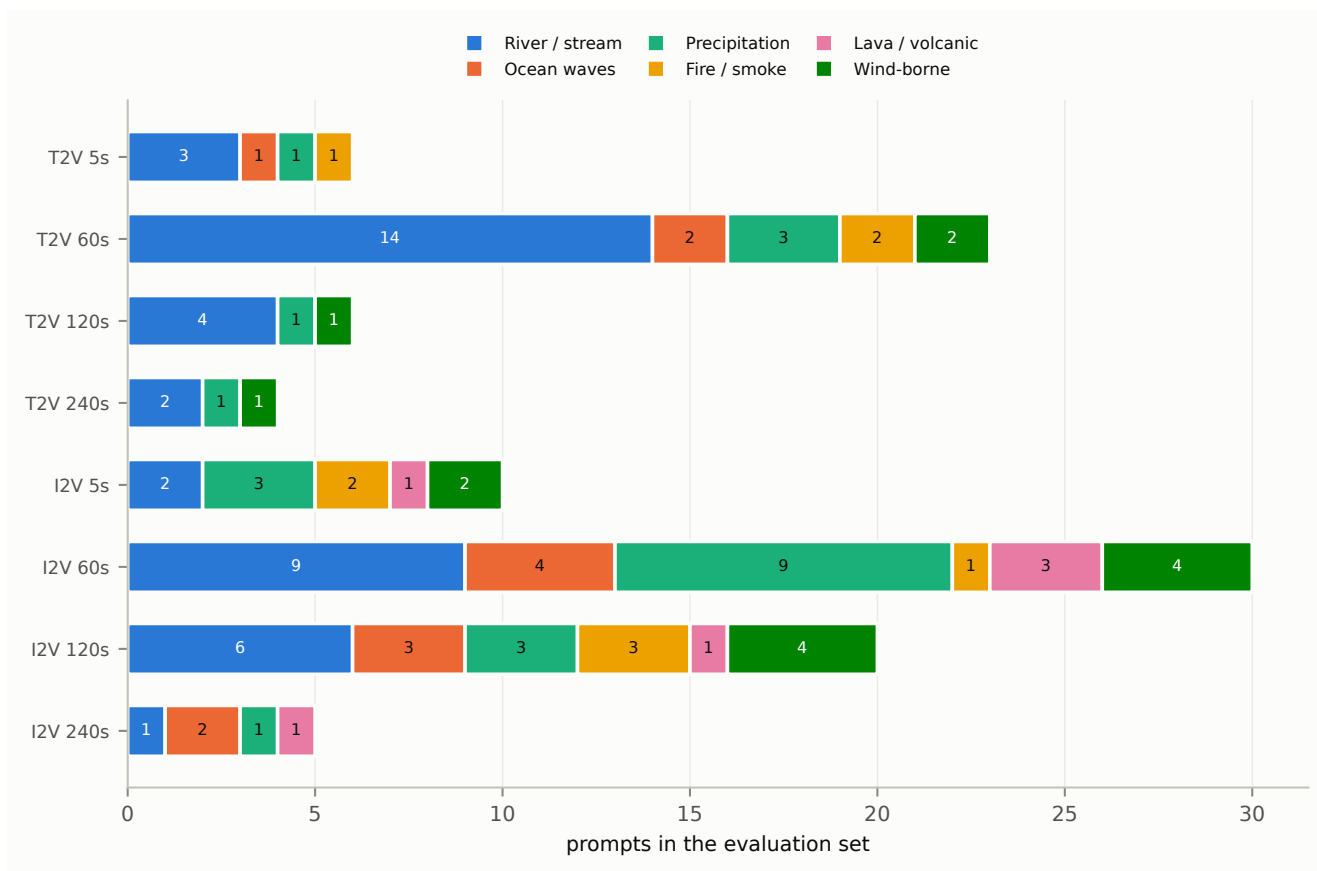


Figure 8. Composition of the evaluation set by scene category, in prompts per category at each track and horizon.

generic metric	fBD	NBF	MCFF	DLR
VB-DD	+0.75	+0.93	+0.89	+0.36
VB-bg	-0.64	-0.71	-0.75	-0.50
VB-smooth / VB-flick	-0.96	-0.82	-0.61	-0.64

Table 8. Rank disagreement at 60 s. Spearman correlation over $n = 7$ method means—descriptive, not inferential. Positive correlation between apparent motion and background flow is the effect the benchmark isolates. Smoothness and flickering rank these systems identically and share a row.

Method	Generic evidence	SNF-Bench evidence	Interpretation
CausVid	VB-DD rank 6/7	NBF rank 3/7, MCFF-L rank 4/7	stable sequence with decaying intended motion
Causal-Forcing	VB-DD rank 1/7	NBF rank 7/7, MCFF-L rank 1/7	substantial apparent motion, strongly contaminated by background displacement
Infinite-Forcing	VB-DD rank 7/7	NBF rank 1/7, MCFF-L rank 7/7	stable sequence with decaying intended motion
LongLive	VB-DD rank 3/7	NBF rank 6/7, MCFF-L rank 3/7	substantial apparent motion, strongly contaminated by background displacement
Reward-Forcing	VB-DD rank 5/7	NBF rank 2/7, MCFF-L rank 6/7	stable sequence with decaying intended motion
Self-Forcing	VB-DD rank 2/7	NBF rank 5/7, MCFF-L rank 2/7	substantial apparent motion, strongly contaminated by background displacement

Table 9. Interpretation changes between generic whole-frame metrics and SNF-Bench at 60s. Neither evaluation is labelled correct; the table identifies information hidden by whole-frame aggregation. Cases are selected by a fixed rank-gap rule applied to all public methods, so none can be selectively omitted.

Checkpoint	horizon	ΔfBD	ΔNBF	$\Delta MCFF-L$	ΔDLR
Causal-Forcing++ (2-step)	5s	+2.47	-0.61	-0.18	-0.01
Causal-Forcing++ (2-step)	60s	-0.76	-0.95	-0.15	-0.08
Causal-Forcing++ (2-step)	120s	-1.71	-8.46	-0.61	-0.17
Causal-Forcing++ (2-step)	240s	+4.06	+0.88	-0.06	-0.28
Causal-Forcing++ (1-step)	60s	-5.05	+1.80	+0.05	-0.20
Causal-Forcing++ (1-step)	120s	-6.34	+2.00	+0.41	-0.14

Table 10. Deployment sensitivity. Change from each checkpoint’s native configuration to the common long-horizon wrapper, for the two checkpoints evaluated both ways. Absolute matched scores are never read as the published method’s performance.

Method	track	dur	n	negative	rate	most negative
CausVid	i2v	120s	20	1	5.0%	-0.055
CausVid	i2v	240s	5	0	0.0%	0.000
CausVid	i2v	5s	10	6	60.0%	-0.450
CausVid	i2v	60s	30	2	6.7%	-0.084
Causal-Forcing++ (1-step)	i2v	120s	20	1	5.0%	-0.012
Causal-Forcing++ (1-step)	i2v	240s	5	0	0.0%	0.175
Causal-Forcing++ (1-step)	i2v	5s	10	0	0.0%	0.041
Causal-Forcing++ (1-step)	i2v	60s	30	0	0.0%	0.013
Causal-Forcing (frame-wise)	i2v	120s	20	3	15.0%	-0.171
Causal-Forcing (frame-wise)	i2v	240s	5	0	0.0%	0.046
Causal-Forcing (frame-wise)	i2v	5s	10	2	20.0%	-0.043
Causal-Forcing (frame-wise)	i2v	60s	30	0	0.0%	0.008
Causal-Forcing++ (2-step)	i2v	120s	20	0	0.0%	0.008
Causal-Forcing++ (2-step)	i2v	240s	5	0	0.0%	0.053
Causal-Forcing++ (2-step)	i2v	5s	10	1	10.0%	-0.102
Causal-Forcing++ (2-step)	i2v	60s	30	0	0.0%	0.005
Causal-Forcing++ (1-step, frame-wise)	i2v	120s	20	1	5.0%	-0.062
Causal-Forcing++ (1-step, frame-wise)	i2v	60s	30	3	10.0%	-0.770
Causal-Forcing++ (2-step, frame-wise)	i2v	120s	20	3	15.0%	-0.192
Causal-Forcing++ (2-step, frame-wise)	i2v	240s	5	2	40.0%	-0.164
Causal-Forcing++ (2-step, frame-wise)	i2v	5s	10	1	10.0%	-0.133
Causal-Forcing++ (2-step, frame-wise)	i2v	60s	30	2	6.7%	-0.014
LTX-Video 13B-0.9.8-distilled	i2v	5s	10	4	40.0%	-0.031
Self-Forcing	i2v	120s	20	3	15.0%	-0.131
Self-Forcing	i2v	240s	5	2	40.0%	-0.012
Self-Forcing	i2v	5s	10	2	20.0%	-0.026
Self-Forcing	i2v	60s	30	8	26.7%	-0.569
Wan2.1-I2V-14B-480P	i2v	5s	10	1	10.0%	-0.242
Wan2.2-I2V-A14B	i2v	5s	10	1	10.0%	-0.490
CausVid	t2v	120s	6	0	0.0%	0.047
CausVid	t2v	240s	4	1	25.0%	-0.384
CausVid	t2v	5s	6	2	33.3%	-0.093
CausVid	t2v	60s	23	0	0.0%	0.005
Causal-Forcing	t2v	120s	6	0	0.0%	0.175
Causal-Forcing	t2v	240s	4	0	0.0%	0.158
Causal-Forcing	t2v	5s	6	2	33.3%	-0.273
Causal-Forcing	t2v	60s	23	3	13.0%	-0.640
Infinite-Forcing	t2v	120s	6	0	0.0%	0.023
Infinite-Forcing	t2v	240s	4	1	25.0%	-0.158
Infinite-Forcing	t2v	5s	6	0	0.0%	0.031
Infinite-Forcing	t2v	60s	23	4	17.4%	-0.359
LongLive	t2v	120s	6	1	16.7%	-1.076
LongLive	t2v	240s	4	0	0.0%	0.022
LongLive	t2v	5s	6	1	16.7%	-0.139
LongLive	t2v	60s	23	2	8.7%	-0.051
Reward-Forcing	t2v	120s	6	1	16.7%	-0.116
Reward-Forcing	t2v	240s	4	0	0.0%	0.002
Reward-Forcing	t2v	5s	12	2	16.7%	-0.145
Reward-Forcing	t2v	60s	23	3	13.0%	-0.317
Rolling-Forcing	t2v	120s	6	3	50.0%	-0.353
Rolling-Forcing	t2v	240s	4	0	0.0%	0.043
Rolling-Forcing	t2v	5s	6	1	16.7%	-0.068
Rolling-Forcing	t2v	60s	24	2	8.3%	-0.038
Self-Forcing	t2v	120s	6	2	33.3%	-0.130
Self-Forcing	t2v	240s	4	1	25.0%	-0.153
Self-Forcing	t2v	5s	6	1	16.7%	-0.046
Self-Forcing	t2v	60s	23	4	17.4%	-1.117

Table 11. Incidence of negative DAR, i.e. clips where global-motion compensation increased measured dynamic-region flow.

category	definition
river_stream	channel water: rivers, streams, creeks, canals, waterfalls, rapids, floods
ocean_waves	open water with wave action: surf, tides, coastal swell, seascapes
precipitation	falling water or ice: rain, snowfall, and the surfaces they wet
fire_smoke	combustion plumes: wildfire, campfire, hearth, urban fire, grill smoke
lava_volcanic	volcanic flow and ejecta: lava channels, ash columns, eruption smoke
windborne	air-carried particulates and wind-driven vegetation: dust, sand, blossom, leaves, thrashing trees, drifting cloud
track/duration	n
i2v/5s	10
i2v/60s	30
i2v/120s	20
i2v/240s	5
t2v/5s	6
t2v/60s	23
t2v/120s	6
t2v/240s	4
track	source
i2v	meta
t2v	keyword
t2v	keyword-body
t2v	override
track	duration
t2v	60s

Table 12. Scene-category balance of the evaluation set, by track and horizon.